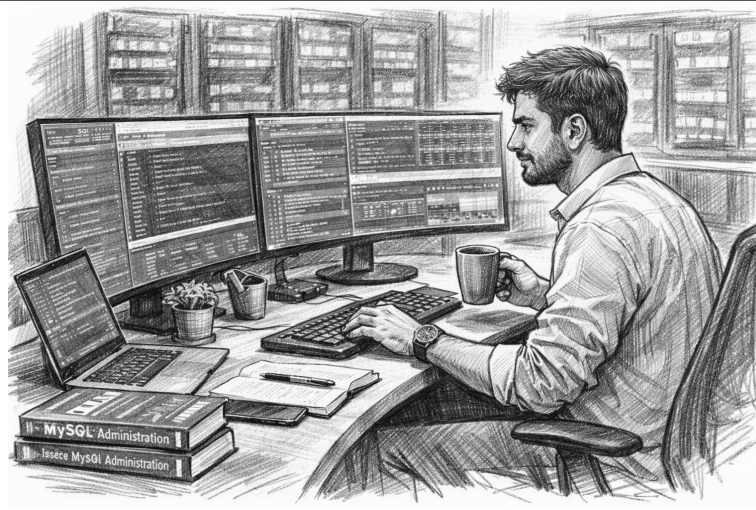




# MySQL Day 25

## Day 25: MySQL Automation & DBA Scripting

In real companies, DBAs cannot manually manage every task. Production environments may have **dozens or hundreds of database servers**, so DBAs automate routine operations.

## Day 25 Objectives

By the end of today you will understand:

- ✓ Why automation is important for DBAs
- ✓ How DBAs automate backups
- ✓ Health-check scripts
- ✓ Using cron jobs
- ✓ Simple shell scripts for MySQL
- ✓ Production automation practices

Backups

Monitoring

Health checks

Log cleanup

Replication checks

Alerts

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## 1 Why Automation is Important

Manual operations cause problems:

| Problem       | Example             |
|---------------|---------------------|
| Human error   | Wrong command       |
| Missed tasks  | Backup not executed |
| Slow response | Issue detected late |

Automation provides:

- ✓ consistency
- ✓ reliability
- ✓ faster operations

These tasks are usually automated using **shell scripts and cron jobs**.

## 2 Common DBA Tasks to Automate

Typical automated tasks:

Daily database backups  
Replication status checks  
Disk usage monitoring  
Slow query log rotation  
Alert notifications  
Database health reports

### 3 Cron Jobs (Task Scheduler)

Linux uses **cron** to schedule tasks.

Open cron configuration:

```
crontab -e
```

```
0 2 * * * /scripts/mysql_backup.sh
```

```
root@secondary:/home/dba#  
root@secondary:/home/dba# crontab -e  
  
0 2 * * * /scripts/mysql_backup.sh
```

Means : Run backup every day at 2 AM

Cront formate:

Minute Hour Day Month Weekday

**Cron**

**Meaning**

0 2 \* \* \*

Daily 2 AM

0 \* \* \* \*

Every hour

\*/5 \* \* \* \*

Every 5 minutes

### Create a mysql\_backup.sh file

```
root@db-01:/usr/local/sbin# sudo nano mysql_backup.sh
```

Note : make sure backup director already created .  
mkdir /backup/mysql

### 4 Automated Backup Script

Example backup script.

```
#!/bin/bash
```

```
DATE=$(date +%F)
```

```
BACKUP_DIR="/backup/mysql"
```

```
mysqldump -u root -pPassword --all-databases >  
$BACKUP_DIR/mysql_backup_$(date +%F).sql
```

```
#!/bin/bash  
  
DATE=$(date +%F)  
BACKUP_DIR="/backup/mysql"  
  
mysqldump -u root -pPassword --all-databases > $BACKUP_DIR/mysql_backup_$(date +%F).sql
```

Save as :mysql\_backup.sh

Make executable:

chmod +x mysql\_backup.sh

```
root@db-01:/usr/local/sbin# chmod +x mysql_backup.sh
root@db-01:/usr/local/sbin#
```

Schedule with cron:

0 2 \* \* \* /scripts/mysql\_backup.sh

```
0 2 * * * /scripts/mysql_backup.sh
```

## 5. Replication Monitoring Script

SHOW REPLICA STATUS\G

see:

Seconds\_Behind\_Master

```
Seconds_Behind_Source: 0
Source_SSL_Verify_Server_Cert: No
Last_IO_Errno: 0
Last_IO_Error:
Last_SQL_Errno: 0
Last_SQL_Error:
```

```
mysql> show replica status\G;
***** 1. row *****
Replica_IO_State: Waiting for source to send event
Source_Host: 172.24.11.82
Source_User: repl_user
Source_Port: 3306
Connect_Retry: 60
Source_Log_File: mysql-bin.000083
Read_Source_Log_Pos: 1730457
Relay_Log_File: OGSRVCMYSQL-01-relay-bin.000235
Relay_Log_Pos: 1730674
Relay_Source_Log_File: mysql-bin.000083
Replica_IO_Running: Yes
Replica_SQL_Running: Yes
```

## Database Health Check Script

```
#!/bin/bash
```

```
echo "Checking MySQL Status"
```

```
mysqladmin ping
```

```
mysql -e "SHOW GLOBAL STATUS LIKE  
'Threads_connected';"
```

```
mysql -e "SHOW GLOBAL STATUS LIKE 'Slow_queries';"
```

## Disk Space Monitoring

if disk space more than 90 % show warning

```
root@db-01:~#  
root@db-01:/# df -h | grep /var/lib/mysql  
root@db-01:/#
```

## Slow Query Log Rotation

if log file can grow large then :

```
mv /var/log/mysql/slow.log /var/log/mysql/slow.log.old  
systemctl restart mysql
```

Run weekly via cron.

```
root@db-01:/usr/local/sbin#  
root@db-01:/usr/local/sbin# sudo nano healthcheck_script.sh
```

```
#!/bin/bash  
  
echo "Checking MySQL Status"  
  
mysqladmin ping  
  
mysql -e "SHOW GLOBAL STATUS LIKE 'Threads_connected';"  
  
mysql -e "SHOW GLOBAL STATUS LIKE 'Slow_queries';"
```

## Automated Alerting

MySQL server down

Replication stopped

Disk space critical

Too many connections

Enterprise tools like:

- Prometheus
- Grafana

provide automated alerts.

## Production Automation Setup

MySQL Server



Health Check Scripts



Monitoring System



Alert System



DBA Notification

Backup Verification Script

ls -lh /backup/mysql

and verify the backup size .

## Basic

- ✓ What is cron?
- ✓ Why automate backups?

## Intermediate

- ✓ How do you monitor replication automatically?
- ✓ How do you schedule scripts in Linux?

## Advanced

- ✓ How would you automate MySQL health checks?
- ✓ How do you verify automated backups?
- ✓ How do monitoring systems alert DBAs?